

2.5

• CASO $n \times n$ / USO LAPLACE

- scegli 3 riga A (quella con 1, 0, 1)

$$A = \begin{pmatrix} 4 & 1 & -1 & 2 \\ 1 & -1 & 1 & -2 \\ 2 & 2 & 3 & 1 \\ 4 & 2 & 3 & 1 \end{pmatrix}$$

LAPLACE

$$\det A = a_{1j} A_{1j} + \dots + a_{in} A_{in}$$

CALCOLO

complementi algebrici:

$$A_{1j} = (-1)^{1+j} \cdot \det \begin{bmatrix} \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots \end{bmatrix}$$

$$0 = A_{31} = (-1)^n \cdot \det \begin{bmatrix} 1 & -1 & 2 \\ -1 & 1 & -2 \\ 2 & 3 & 1 \end{bmatrix}$$

$$35 = A_{32} = (-1)^5 \cdot \det \begin{bmatrix} 4 & -1 & 2 \\ 1 & 1 & -2 \\ 4 & 3 & 1 \end{bmatrix}$$

$$15 = A_{33} = (-1)^6 \cdot \det \begin{bmatrix} 4 & 1 & 2 \\ 1 & -1 & -2 \\ 4 & 2 & 1 \end{bmatrix}$$

$$\det A = 35 \cdot 1 + 15 \cdot 2 = 65$$